

GABRIEL POIRIER

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PROFESSIONAL SUMMARY

Software & Machine Learning Engineer graduating from **ENSIMAG** with a strong background in computer science, low-level systems, and deep learning pipelines. Experienced in PyTorch architectures, explainable AI, signal processing, and cloud/observability tools. Passionate about solving complex algorithmic challenges and building scalable software.

WORK EXPERIENCE

INRAE

Data Science & Machine Learning Engineering Intern

Grenoble, France

April 2026 – September 2026

- Designed and evaluated 1D Convolutional Neural Networks (CNNs) and dictionary learning pipelines in **PyTorch** for complex signal classification.
- Addressed domain shift challenges between simulated and experimental datasets using quantile normalization and fine-tuning strategies.
- Integrated Explainable AI (XAI) frameworks (**Grad-CAM**, attribution maps) to interpret model decisions and identify spectral features.
- Optimized hyperparameter tuning via **Optuna** and validated cross-validation performance.

Nomadia Group (70PTEAM)

Software & Systems Engineer Intern

Grenoble, France

June 2025 – August 2025

- Migrated internal tool authentication from Microsoft OAuth2 to Google OAuth2, ensuring seamless SSO integration.
- Implemented automated backup server infrastructure and containerized backend services using **Docker**.
- Built **Grafana** monitoring dashboards and configured automated alert pipelines for production performance tracking.

EDUCATION

ENSIMAG - Grenoble INP

Master of Science in Engineering - Computer Science & Applied Mathematics

Grenoble, France

Sep 2023 – Aug 2026

- **Specialization:** Information Systems Engineering (ISI).
- **Coursework:** Low-Level Architecture, Operating Systems, Networks, Advanced Algorithms, Language Theory, Cryptography.
- **Erasmus Exchange:** Lund University, Sweden.

Lycée Clemenceau

CPGE PC (Preparatory Classes – Mathematics, Physics & Chemistry)*

Nantes, France

Sep 2021 – Jun 2023

TECHNICAL PROJECTS

Signal Processing & ML Transfer Learning Pipeline

Python, PyTorch, SciPy

- Built an end-to-end framework for zero-shot evaluation, regularized non-negative least squares (NNLS), and transfer learning.

Compiler & Systems Project (ENSIMAG)

C, Java, Assembly

- Developed a compiler targeting pseudo-assembly code with full parsing, lexical analysis, and code generation.

TECHNICAL SKILLS & INTERESTS

- **Languages:** Python, C, C++, Java, SQL, JavaScript, Assembly (x86), Bash.
- **Machine Learning & Data:** PyTorch, NumPy, SciPy, Scikit-Learn, Grad-CAM, Optuna, Signal Processing.
- **Systems & DevOps:** Linux, Docker, Grafana, OAuth2, Git, CI/CD, PostgreSQL.
- **Interests:** Cybersecurity (CTF competitions on Root-Me), Guitar, Basketball, Snowboard.